MIL-A-6752A(ASG)
24 DECEMBER 1953
Superseding
MIL-A-6752
14 June 1950
MIL-A-5125(USAF)
9 December 1949

MILITARY SPECIFICATION

AMMETERS, VOLTMETERS, AND LOADMETERS; DIRECT CURRENT

This specification has been approved by the Department of the Air Force and by the Navy Bureau of Aeronautics.

1. SCOPE

1.1 This specification covers the general requirements of d-c voltmeters, and of 50-millivolt d-c ammeters and loadmeters used in military aircraft.

2. APPLICABLE DOCUMENTS

2.1 The following specifications, standards, drawings, and publications, of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

Military

MIL-D-5028	Drawings, and Data Lists; Preparation of (For Engines, Accessories, and Other Auxiliary Equipment)
MIL-P-6064	Packaging of Lightweight Aircraft Accessories
MIL-S-7742	Screw Threads, Standard, Aeronautical
JAN-S-44	Shock-Testing-Mechanism for Electrical-Indicating Instruments (2-1/2 and 3-1/2 Inch, Round, Flush-Mounting, Panel-Type)

Air Force-Navy Aeronautical

AN-L-1	Luminescent Material; Fluorescent

STANDARDS

MIL-STD-129	Marking of Shipments
MIL-STD-130	Identification Marking of U. S. Military Property
MS25025	Loadmeter, Direct Current, Type J-1

DRAWINGS

Air Force-Navy Aeronautical Standard Drawings

AN3200	Shunt-Ammeter, External, 50 Millivolt
AN3201	Ammeter - Direct Current
AN3203	Voltmeter - Direct Current, Panel
AND10400	Numerals and Letters - Aircraft Instrument Dial, Standard Form of

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T07:24Z
Check the source to verify that this is the current version before use.

PUBLICATIONS

Air Force-Navy Aeronautical Bulletin

No. 143	Specifications and Standards; Use of
No. 405	Storage Life - Aeronautical Article

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Qualification.- The meters furnished under this specification shall be a product which has been tested and has passed the Qualification tests specified herein.

3.2 Materials.- Materials shall conform to applicable specifications as specified herein. Materials which are not covered by applicable specifications shall be of the best commercial quality and suitable for the purpose intended.

3.2.1 Selection of materials.- Specifications and standards for all materials, parts, and Government certification and approval of processes and equipment, which are not specifically designated herein and which are necessary for the execution of this specification, shall be selected in accordance with ANA Bulletin No. 143, except as provided in the following paragraph.

3.2.1.1 Standard parts.- Standard parts (MS, AN, or JAN) shall be used wherever they are suitable for the purpose, and shall be identified on the drawing by their part numbers. Commercial utility parts such as screws, bolts, nuts, cotter pins, etc, may be used, provided they possess suitable properties and are replaceable by the standard parts (MS, AN, or JAN) without alteration, and provided the corresponding standard part numbers are referenced in the parts list and, if practicable, on the contractor's drawings. In the event there is no suitable corresponding standard part in effect on date of invitation for bids, commercial parts may be used provided they conform to all requirements of this specification.

3.3 Design and construction.- The meter shall be so designed and constructed that no parts will work loose in service.

3.3.1 Case.- Meter cases shall conform to Drawings AN3201, AN3203, or Standard MS25025, as applicable. The body of the case shall be made either of synthetic material, composed of a suitable filler and phenol condensation binder, or of metal. If of synthetic material, it shall be of the hot-molded type, and shall be uniform in texture, and shall have a smooth surface. The portion of the face exposed to view from the front of the panel shall have a durable dull black finish.

3.3.2 Cover glass.- The cover glass shall be clear, and free from flaws that will seriously affect the readability of the meters and shall have a smooth edge and shall not be beveled.

3.3.3 Dial.- The dial shall be made of metal or of paper and shall be marked as specified herein. Paper dials will be satisfactory only when adequately bonded to and backed by a metal support and when they meet all tests specified herein without fading, discoloring, warping, or other deterioration.

3.3.4 Pointers.- The body of the pointer shall be a maximum of 1/16 inch wide. The entire visible portion of the pointer shall be marked with fluorescent-luminescent material conforming to Specification AN-L-1.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T07:24Z
Check the source to verify that this is the current version before use.

3.3.5 Scale length.-

3.3.5.1 The scale length of ammeters and voltmeters shall be not less than 1-7/16 inches as measured by the arc described by the pointer tip.

3.3.5.2 The scale length of loadmeters shall be not less than 1 inch as measured by the arc described by the pointer tip.

3.3.6 Scale visibility.- The scale and transparent front shall be so designed that the full scale of the meter can be easily read at any angle up to 45 degrees below or 30 degrees on either side of the axis through the center of the meter perpendicular to the plane of mounting with illumination coming from above and in front and at an angle of 45 degrees from the same axis. Under these conditions there shall be no objectional shadows on the graduated scale.

3.3.7 Parallax.- To reduce parallax, the clearance between pointer and the graduated scale shall not exceed 0.05 inch.

3.3.8 Polarity.- The polarity of the meter terminals shall be legibly and permanently marked. For all meters, the left-hand terminal (as viewed from the rear of the meter in the normal mounted position) shall be the positive terminal and shall be marked "+" on or near the terminal.

3.3.9 Bearings and pivots.- Moving elements shall turn on hardened, accurately ground, and highly polished steel pivots bearing on highly polished first-quality sapphire, diamond jewels, or other materials that have been approved as an alternate to precious gems.

3.3.10 Means of adjustment.- It shall be ascertained by suitable tests that the meters have some practical shop means of adjustment to accuracy. Charging of the magnet and magnetically treating to accuracy shall be considered as meeting this requirement. Internal resistances shall be easily accessible.

3.3.11 Pointer zero correction adjustment.- All meters having spring control shall have a zero correction adjustment accessible from the front of the case. It shall provide for a range of adjustment, above and below the zero point, of not less than 3 percent of the scale length. The zero adjustment shall be suitably insulated to eliminate all danger to the operator. The adjustment mechanism (or screw) shall be designed and constructed to prevent damage when meter is placed face down and also shall have sufficient friction to prevent it from shaking loose and changing its adjustment due to the Vibration tests specified herein. The range of zero adjustment shall be taken and recorded.

3.3.12 Bearing adjustment screw.- To prevent tampering, the bearing adjustment screw shall not be exposed.

3.3.13 Terminal posts.- The terminal posts shall be threaded and shall be provided with the proper nuts and lock washers for attaching cable terminals, in accordance with Drawings AN3201, AN3203, and Standard MS25025. Plating of the posts shall be subject to the approval of the procuring activity.

3.3.14 Adjustment.- The ammeters and loadmeters shall be adjusted to provide full-scale reading at 50 millivolts with a shunt lead resistance of 0.5 ohm in series with the meter. The shunt shall conform to Drawing AN3200.

3.4 Interchangeability.- All parts having the same manufacturer's part number shall be directly and completely interchangeable with each other with respect to installation and performance. Changes in manufacturer's part numbers shall be governed by the drawing number requirements of Specification MIL-D-5028.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T07:24Z
Check the source to verify that this is the current version before use.

3.5 Screw threads.- Screw threads with major diameters of 0.060 inch or more shall conform to Specification MIL-S-7742.

3.6 Operational range.- The range of operation for the meters shall be from -55° to +70°C (-67° to +158°F).

3.7 Damping.- The meters shall have an efficient damping arrangement. The damping factor shall be not less than the following:

(a)	Loadmeters	1.5
(b)	Ammeters	2.5
(c)	Voltmeters	2.0

3.8 Balancing.- When spring-controlled instruments are tilted 60 degrees from the normal vertical position, in any direction, the pointer tip shall remain within 2 percent of total scale length of the zero point. Elements free to swing shall show a good mechanical balance. The balance shall be easily and readily adjustable.

3.9 Temperature influence.- The temperature influence of the meters when tested as specified herein shall not exceed the following.

(a)	Loadmeters	1.0
(b)	Ammeters	1.0
(c)	Voltmeters	0.50

3.10 Accuracy.-

3.10.1 Room temperature.- Unless otherwise specified on the applicable drawing, the meters shall indicate within ±2 percent of full-scale deflection at room temperature of 25° ±5°C (77° ±9°F).

3.10.2 Low temperature.- The change in reading when the meter is submitted to test at -55°C (-67°F) shall not exceed 8 percent for loadmeters and ammeters and 4 percent for voltmeters.

3.10.3 High temperature.- The change in reading when the meter is subjected to High temperature test as specified in Section 4 shall not exceed 8 percent for loadmeters and ammeters and 4 percent for voltmeters.

3.11 Response time.- Response time shall not exceed 2.5 seconds when a voltage or current sufficient to give full-scale momentary deflection is suddenly applied.

3.12 Magnetic shielding.- The meters shall be magnetically shielded so that they will not cause a deflection of the compass of more than 6 degrees when tested as specified in Section 4.

3.13 Insulation resistance.-

3.13.1 An alternating current, commercial frequency, voltage as specified below, applied for a period of 1 minute shall cause no breakdown of insulation:

(a)	Loadmeters	110 to 120 V rms
(b)	Ammeters	1,000V rms
(c)	Voltmeters	1,000V rms

3.13.2 The insulation resistance between all live parts and the metal panel upon which the instrument is mounted shall be not less than 20 megohms at a potential of 500 volts dc.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T07:24Z
Check the source to verify that this is the current version before use.

3.14 Humidity resistance.- The meters shall operate satisfactorily after subjection to Humidity test as specified in Section 4. The permanent change in indication after the second humidity cycle shall not exceed 3 percent.

3.15 Overload capacity.-

3.15.1 Momentary overload.- Meters shall be sufficiently rugged to withstand a series of momentary overloads as described in Section 4 without incurring damage other than a permanent error, at any part of the scale, of not more than 1 percent of the scale length.

3.15.2 Sustained load.- Meters shall withstand for 8 hours sustained loads as specified in Section 4 without a permanent change in indication of more than 1 percent of the scale length, without a permanent zero shift greater than 0.5 percent of the scale length, and without any other permanent damage. The temporary zero shift shall not exceed 1 percent of the scale length.

3.16 Shock resistance.- Permanent errors in the meter indication resulting from shock of the character and intensity specified herein, shall not exceed 2 percent of full scale deflection. No screws, bearings, pivots, or other parts shall become loosened or unduly damaged so as to affect the operation of the meter.

3.17 Vibration.- Permanent errors in indication due to vibration of the character and intensity specified in Section 4 shall not exceed 2 percent of full-scale deflection. The friction error on the basis of the difference between ascending and descending readings shall not exceed 4 percent of full-scale deflection. No screws, bearings, pivots or other parts shall become loosened or unduly damaged so as to affect the operation of the meter.

3.18 Mounting nuts.- Unless otherwise specified, each mounting lug shall be fitted with a self-locking type, threaded insert nut, so attached as to prevent loosening under an axial load of 25 pounds and to prevent rotation by more than a fraction of a turn under a torque of 10 inch-pounds.

3.19 Mounting lugs.- Mounting lugs of the complete assembled case shall withstand for 1 minute, without fracture, a load of 175 pounds applied along the insert nut axis and toward the front of the meter.

3.20 Dimensions.- The meters shall conform to the dimensions shown on Drawings AN3201, AN3203, and Standard MS25025.

3.20.1 Weight.- The weight (lb) shall not exceed the following values:

(a)	Loadmeters	0.35
(b)	Ammeters	1.2
(c)	Voltmeters	1.0

3.21 Markings.- All markings shall be durable to withstand usage encountered in service.

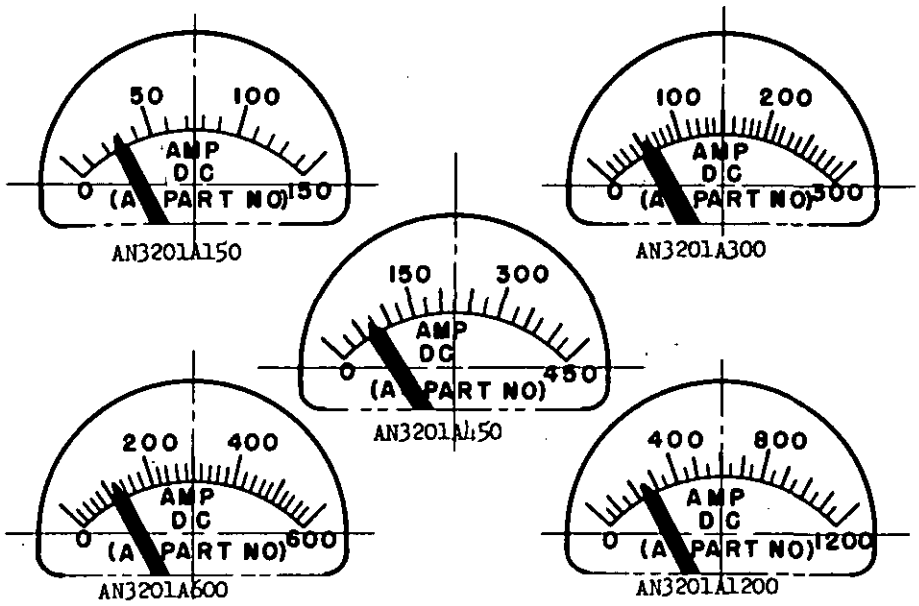
3.21.1 Dial markings.- Dials shall be marked as shown on figures 1, 2, and 3. The form of the letters and numerals shall conform to Drawing AND10600. Numerals shall distinctly indicate the graduations to which each applies. When several numerals are used in one group, the space between numerals shall be not less than 1/64 inch.

3.21.1.1 Fluorescent-luminescent material.- Luminescent material shall conform to Specification AN-L-1.

3.21.2 The following note in letters 5/32 inch high shall be marked on the back of each ammeter and loadmeter:

"USE WITH EXTERNAL SHUNT"

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T07:24Z
Check the source to verify that this is the current version before use.

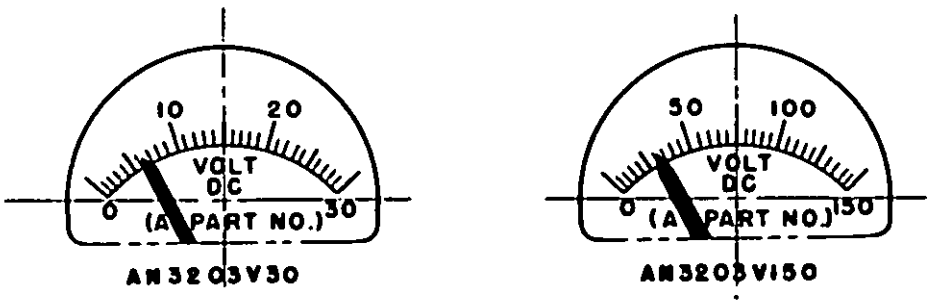


OVERLAP OF POINTERS TO BE 50 ±15 PERCENT OF MINOR GRADUATIONS

MARKING	HEIGHT OR LENGTH, INCHES ±0.010	WIDTH OF LINE OR GRADUATION INCHES ±0.005	FINISH
ALL NUMERALS LETTERING (EXCEPT DC) DC MAJOR GRADS INTERMEDIATE GRADS SECTOR LINE	0.109 0.156 0.094 0.188 0.156 - - -	- - - - - - - - - - - - 0.030 0.025 0.015	FLUORESCENT-LUMINESCENT MATERIAL
MINOR GRADS	0.125	0.015	DURABLE DULL WHITE OR NATURAL METAL
AN PART NO	0.062	- - - -	DURABLE DULL BLACK

FIGURE 1. Ammeter dials and markings

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T07:24Z
Check the source to verify that this is the current version before use.

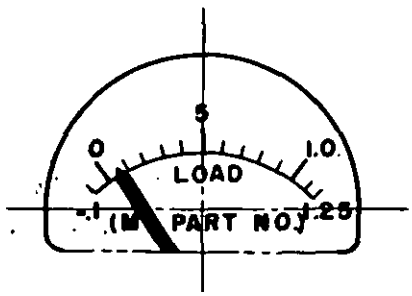


OVERLAP OF POINTERS TO BE 50 ±15 PERCENT OF MINOR GRADUATIONS

MARKINGS	HEIGHT OR LENGTH, INCH ±0.010	WIDTH OF LINE OR GRADUATION INCH ±0.005	FINISH
ALL NUMERALS	0.109	- - - -	FLUORESCENT-LUMINESCENT MATERIAL
LETTERING (EXCEPT DC)	0.156	- - - -	
DC	0.094	- - - -	
MAJOR GRADS	0.188	0.030	
INTERMEDIATE GRADS (EXCEPT 5 AND 15 OF AN3203V30)	0.156	0.025	
MINOR GRADS, 21 THRU 29 OF AN3203V30.	0.125	0.015	
SECTOR LINE	- - - -	0.015	
INTERMEDIATE GRADS 5 AND 15 OF AN3203V30	0.156	0.025	DURABLE DULL WHITE OR NATURAL METAL
MINOR GRADS	0.125	0.015	
AN PART NO.	0.062	- - - -	DURABLE DULL BLACK

FIGURE 2. Voltmeter dials and marking

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T07:24Z
Check the source to verify that this is the current version before use.



OVERLAP OF POINTER TO BE 50 ±15 PERCENT OF MINOR GRADUATIONS

MARKING	HEIGHT OR LENGTH INCH ±0.010	WIDTH OF LINE OR GRADUATION INCH ±0.005	FINISH
ALL NUMERALS LOAD MAJOR GRADS INTERMEDIATE GRADS	0.094 0.094 0.094 0.062	0.015 0.015 0.015 0.015	FLUORESCENT-LUMINESCENT MATERIAL
MINOR GRADS	0.031	0.015	DURABLE DULL WHITE
LETTERING (EXCEPT LOAD)	0.062	- - - -	DURABLE DULL BLACK

FIGURE 3. Loadmeter dial and marking

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T07:24Z
Check the source to verify that this is the current version before use.

3.22 Identification of product.- The physical item shall be marked for identification in accordance with Standard MIL-STD-130. The nameplate shall contain the information presented below:

Name (as applicable)
AN or MS Part No.
Specification MIL-A-6752A
Stock No.
Manufacturer's Part No.
Manufacturer's Serial No.
Contract or Order No.
Manufacturer's Name or trade-mark
US

3.22.1 Use of AN or MIL designations.- AN or MIL designations shall not be applied to a product, except for Qualification test samples, nor referred to in correspondence, until notice of approval has been received from the activity responsible for qualification, or from the Aeronautical Standards Group.

3.23 Workmanship.-

3.23.1 General.- The meter, including all parts and accessories, shall be constructed and finished in a thoroughly workmanlike manner. Particular attention shall be given to neatness and thoroughness of soldering, wiring, marking of parts and assemblies, welding and brazing, painting, riveting, machine-screw assemblies, and freedom of parts from burrs and sharp edges.

3.23.2 Dimensions.- Dimensions and tolerances not specified shall be as close as is consistent with the best shop practices. Where dimensions and tolerances may affect the interchangeability, operation, or performance of the meter, they shall be held or limited accordingly.

3.23.3 Screw assemblies.- Assembly screws and bolts shall be tight. The word "tight" means that the screw or bolt cannot be appreciably tightened further, without damage or injury to the screw or bolt or threads.

3.23.4 Riveting.- Riveting operations shall be carefully performed to insure that the rivets are tight and satisfactorily headed.

3.23.5 Cleaning.- The meter shall be thoroughly cleaned of loose, spattered, or excess solder, metal chips, and other foreign material, after final assembly. Burrs and sharp edges, as well as resin flash that may crumble, shall be removed.

4. QUALITY ASSURANCE PROVISIONS

4.1 Classification of tests.- The inspection and testing of meters shall be classified as follows:

(a) Qualification tests: Qualification tests those tests performed on samples submitted for approval as qualified products.

(b) Acceptance tests: Acceptance tests are those tests performed on individual lots which have been submitted for acceptance.

4.2 Qualification tests.-

4.2.1 Sampling instructions.- The Qualification test samples shall consist of eight meters of a single range upon which qualification is desired. If these meters of

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T07:24Z
Check the source to verify that this is the current version before use.

a single range satisfactorily pass the Qualification tests specified herein, meters of the same construction but other ranges may be recommended for type approval and inclusion in the approved list without further test. One assembly drawing or drawings showing detail parts shall be furnished with Qualification test samples. Samples shall be forwarded to the activity designated in the letter of authorization from that activity (see paragraph 6.4), plainly identified by securely attached durable tags marked with the following information:

Samples for Qualification Tests
(Name of meter)
AN or MS Part No.
Manufacturer's Part No.
Name of manufacturer
Submitted by (name) (date) for Qualification test in accordance with requirements of Specification MIL-A-6752A reference authorizing letter).

4.2.2 Tests.- Qualification test samples, as specified below, shall be subjected to all the tests of this specification described under "Test methods," conducted in the order listed below. At the option of the activity responsible for qualification, the Qualification tests may be supplemented with tests under actual service conditions:

Test group 1 (All samples)

(a) Examination of product
(b) Zero adjustment
(c) Balancing
(d) Accuracy - room temperature
(e) Damping
(f) Response time
(g) Magnetic shielding
(h) Temperature influence
(i) Accuracy - high temperature
(j) Accuracy - low temperature exposure
(k) Insulation resistance

Test group 2 (Two samples)

(a) Overload
    (1) Momentary
    (2) Sustained
(b) Mounting lugs
(c) Mounting nuts

Test group 3 (Two samples)

(a) Humidity conditioning
(b) Humidity cycling

Test group 4 (Two samples)

(a) Shock
(b) Vibration

Test group 5 (Two samples)

Supplementary tests, if necessary

4.3 Acceptance tests.- The Acceptance tests shall consist of Individual tests, Sampling tests, and Special tests.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T07:24Z
Check the source to verify that this is the current version before use.

4.3.1 Individual tests.- Each meter submitted for acceptance under contract shall be subjected to the following tests, as described under "Test methods" of this specification:

(a) Examination of product
(b) Zero adjustment
(c) Balancing
(d) Accuracy - room temperature

4.3.2 Sampling tests.- Five meters which have been manufactured under essentially the same conditions shall be selected at random from each lot of 500 or fraction thereof on the order and subjected to the following tests as described under "Test methods":

(a) Response time
(b) Magnetic shielding
(c) Vibration
(d) Insulation resistance

4.3.3 Special tests.- Three of the meters of each part number representing the manufacturer's contract or order, and samples selected as often thereafter as the procuring activity deems advisable, shall be subjected to any or all of the tests listed below as described under "Test methods":

(a) Humidity resistance
(b) Low temperature exposure
(c) Mounting nuts
(d) Mounting lugs

4.3.4 Rejection and retest.-

4.3.4.1 When tests are specified on a number of items that are selected as representative of a certain lot, and one or more of this number fails to meet the specified test(s), acceptance of all items shall be withheld until the extent and cause of failure is determined. For operational reasons, Individual tests may be continued pending investigation of a Sampling test failure, but the final acceptance of the product is contingent upon the Inspector's decision regarding the over-all conformance of the product to specification requirements. If investigation indicates that the defects may exist on items previously accepted, full particulars concerning the defect(s) found, including recommendations for correction, shall be furnished to the procuring activity.

4.3.4.2 Items which have been rejected may be reworked or replaced to correct the defects, and resubmitted for acceptance. Before resubmitting, full particulars concerning previous rejection and the action taken to correct the defects found in the original shall be furnished the Inspector. Units rejected after retest shall not be resubmitted without the specific approval of the procuring activity.

4.4 Test conditions.-

4.4.1 Unless otherwise specified, the meters shall be tested at room temperature of 25° ±5°C (77° ±9°F).

4.4.2 Tapping.- Either the meter or its support shall be tapped lightly in taking meter readings, except when ascertaining the effect of friction after vibration.

4.5 Test methods.-

4.5.1 Examination of product.- Each meter submitted for acceptance under contract shall be examined to determine conformance with this specification with respect to material, workmanship, and marking.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T07:24Z
Check the source to verify that this is the current version before use.

4.5.2 Damping.- Voltage or current sufficient to produce momentary full scale deflection shall be applied to the meter. The maximum momentary deflection shall be taken and recorded. After the pointer has come to rest, the steady deflection shall be taken and recorded. The damping factor shall be considered the ratio of the steady deflection in angular degrees to the difference between the maximum momentary deflection and the steady deflection in angular degrees.

4.5.3 Insulation resistance.- The voltage as specified in paragraph 3.13 shall be applied between each terminal post and the metal panel upon which the instrumnt is mounted.

4.5.4 Balancing.- The pointer shall be set to the center of the zero scale division with the meter in the vertical position. The instrument shall then be tilted 60 degrees from the vertical in any direction, and the position of the pointer shall not deflect more than the amount specified in Section 3.

4.5.5 Temperature influence.- Meters shall be tested over a range of 20°C above and 20°C below 25°C. Readings shall be taken at not less than five points (approximately equidistant) on the scale, including maximum scale value and the temperature influence computed.

4.5.5.1 The temperature influence shall be considered 1/2 the percentage change in indication caused by the 20° departures from 25°C. If the influences above and below 25°C are not equal, the greater value shall be considered the temperature influence ascertained.

4.5.6 Accuracy.-

4.5.6.1 Room temperature.- Each meter shall be calibrated to determine the accuracy of the instrument at room temperature.

4.5.6.2 High temperature.- Meters shall be placed in heated air and maintained at a temperature of 70°C for 16 hours. At the conclusion of this period and while still heated to this temperature, meters shall be read throughout the calibrated scale to determine the accuracy at high temperatures. It shall be observed whether or not the meter indicates freely at all parts of the scale. Meters shall be allowed to cool at 25°C for 8 hours and another set of readings taken throughout the calibrated scale. The tests shall be made three times successively.

4.5.6.2.1 Upon completion of the above three tests, meters shall be exposed to a temperature of +93°C for a period of 48 hours. These meters shall then be allowed to cool at room temperature and tested to determine that no damage has resulted to the meter.

4.5.6.3 Low temperature exposure.- The meter shall be subjected for a minimum period of 48 hours to a temperature of -65°C (-85°F). The temperature shall then be raised to a temperature of -55°C (-67°F) and tested as specified for the Accuracy test at low temperature to determine that no damage has resulted to the instrument.

4.5.7 Response time.- Voltage or current sufficient to produce momentary full-scale deflection shall be applied to the meter. The length of time, in seconds, required for the pointer to come to rest shall be noted. This test shall be repeated several times and the average value taken as the response time of the meter. The pointer is to be considered as having come to apparent rest when it has reached the actual rest point within ±1 percent of the scale length.

4.5.8 Magnetic shielding.- The meter shall be rotated in a vertical plane about a short bar magnet-type compass with the nearest part of the meter 5 inches from and magnetically east or west of the center of the compass. Starting directly under the compass, the meter shall be held in position 0°, 45°, 90°, 135°, 180°, 225°, 270°, and 315° from the initial position. At each of these positions the meter shall be in its normal upright mounting position. The horizontal magnetic field intensity shall be 0.17 to 0.19 gauss.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T07:24Z
Check the source to verify that this is the current version before use.

4.5.9 Zero adjustment.- Each meter shall be examined to determine that the pointer is set at the center of the zero scale division with no excitation applied to the meter.

4.5.10 Humidity resistance.-

4.5.10.1 Humidity conditioning.- One meter with the cover in place and mounted in a vertical position shall be subjected to the following Humidity cycling tests. At intervals during the test, the meter should be examined for collection of moisture in the interior of the case, corrosion of metal parts, loosening or deterioration of the scale in any way, etc. The second meter shall have its cover removed and shall be subjected to the Humidity cycling tests, and inspection for the same features shall be conducted with the exception of collection of moisture within the case. The resistance of the internal multipliers, where used, shall be measured in each of the tests specified above.

4.5.10.2 Humidity cycling tests.- The meters shall be placed in a chamber and kept for 6 hours at a temperature of 65°C (149°F), and relative humidity of 95 percent. After the required time, the meters shall be cooled to normal ambient temperature for 18 hours in this atmosphere, taking precaution to prevent dripping on the meters, and then tested for response, accuracy, damping factor, and balance. The complete procedure shall be repeated a second time. The meters shall operate satisfactorily at the conclusion of these tests.

4.5.11 Overload capacity.-

4.5.11.1 Momentary overload.- The mechanism of meters shall be subjected to the following series of momentary overloads:

(a) A load of 10 times full-scale value, applied 9 times for a duration of 1/2 second each, with a 1-minute interval between the successive applications, then followed by one application for a 5-second period (making a total of 10 applications during an elapsed period of 9 minutes 9-1/2 seconds, during which time current was applied for a total of 9-1/2 seconds).

(b) Voltmeters shall withstand momentary overload of three times rated voltage applied in the manner as specified above. The pointer shall be reset to zero and the calibration checked over the length of the scale.

4.5.11.2 Sustained overload.- Ammeters and loadmeters shall withstand an applied load corresponding to 20 percent more than rated capacity, and voltmeters shall withstand an overload of 3 times rated voltage for a period of 8 hours. The temporary zero shift of each meter shall be noted immediately upon withdrawal of the 8-hour period. Permanent errors in the indication and the permanent zero shift shall be ascertained 16 hours after removal of load.

4.5.12 Shock.- The shock testing mechanism complying with Specification JAN-S-44 is suitable for conducting this test. Meters shall be mounted in their normal vertical mounting position and subjected to 10 shocks with an acceleration value of 50 gravity units each in a direction normal to the axis of the moving system at a rate of build-up and decay as defined by the shock-test device specified. The meters shall then be rotated 90 degrees counterclockwise in the plane of mounting and subjected to 10 shocks of the same magnitude and direction as previously. The meters shall then be mounted in a horizontal plane, face upward, and subjected to 10 shocks of the same magnitude as at first but in a downward direction parallel to the axis of the moving system. Permanent error in indication shall be ascertained. Meter mechanisms shall be examined for loosening of or damage to parts. At the conclusion of this Shock test, the meters shall be subjected to the Vibration tests specified herein.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T07:24Z
Check the source to verify that this is the current version before use.

4.5.13 Vibration.- Meters shall be subjected to a vibration of 24 hours consisting of 8-hour periods in each of the following three positions:

Position 1 - Normal vertical mounting position.

Position 2 - Rotated 90 degrees counterclockwise in plane of mounting.

Position 3 - Horizontal plane with face upward.

The construction of the vibrating platform shall be such as to give simple harmonic motion at a frequency of 400 cpm and with an amplitude of 1/16 inch at the horizontal center line of the meter.

4.5.13.1 Meters shall also be mounted on a vibration stand giving a frequency between 500 and 2,500 vibrations per minute of such a character that a point of the meter case will describe, in a plane inclined 45 degrees to the plane of the face, a circle with a diameter of not less than 0.018 inch or more than 0.020 inch. The meter shall be vibrated for a total of 2 hours at any desired frequency between 500 to 2,500 vibrations per minute, but shall not be vibrated any one frequency for more than 30 minutes. The meter shall be successively deflected to approximately five equidistant points on the scale during this test. In no case, however, shall the pointer be held at any given scale mark for more than 30 minutes.

4.5.13.2 At the conclusion of the Vibration tests, the meter shall be connected to a source of potential or current, as applicable, and readings taken throughout the calibrated scale. These readings are to be taken on both ascending and descending scales, the latter to observe the effect of friction. The errors due to vibration shall not exceed the values specified herein. The zero adjustment of the meter shall not have changed as a result of this test.

4.5.13.3 The condition of the pivots and bearings after the Vibration tests shall be observed through a microscope of at least 40 diameters.

4.5.14 Mounting nuts.- Each mounting nut shall have the specified load applied in such a manner as to tend to pull the nut out of the lug toward the rear of the case. Each insert nut shall be subjected to a torque of 10 inch-pounds about the thread axis.

4.5.15 Mounting lugs.- The instrument case shall be mounted face downward on the movable head of a suitable testing machine with the diameter of the case in a horizontal plane in order that the mounting lugs receive no added support. A suitable pin shall be inserted through the hole in the mounting nut and attached to the pull strap in the stationary head of the machine. The specified load shall be applied to each lug for a period of 1 minute.

5. PREPARATION FOR DELIVERY

5.1 Application.- The requirements of Section 5 apply only to direct purchases by or direct shipments to the Government.

5.2 Packaging.- The meter shall be packaged in accordance with Specification MIL-P-6064, group VI, in a type II, class I container.

5.3 Packing.- The interior packages shall be packed for domestic or overseas shipment, as specified, in accordance with applicable requirements of Specification MIL-P-6064.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T07:24Z
Check the source to verify that this is the current version before use.

5.4 Marking of shipments.- Interior packages and exterior shipping containers shall be marked in accordance with Standard MIL-STD-129. The nomenclature shall be as follows: Ammeter, Direct Current, Panel Mounted, Marked Amperes (Range of Inscription) Range Amperes ___ (As applicable) Round Square Flange Mounting, Flange 2-3/8 x 2-3/8 x 3/8 Inch Thick, AN Part No. AN3201- , (as applicable), Specification MIL-A-6752A.

or

Voltmeter, Direct Current, Panel Mounted, Marked Volt (Range of Inscription) Range Volts ____, (as applicable) Round Square Flange Mounting, Flange 2-3/8 x 2-3/8 x 3/8 Inch Thick, AN Part No. AN3203- , (as applicable) Specification MIL-A-6752A.

or

Loadmeter, Direct Current, Panel Mounted, Marked Load (Range of Inscription) Range Load ____, (as applicable) Round Square Flange Mounting, Flange 1-3/4 x 1-3/4 x 5/16 Inch Thick, MS Part No. ____, (as applicable) Specification MIL-A-6752A.

5.4.1 Reinspection dates.- Reinspection marking shall be added in accordance with ANA Bulletin No. 405.

6. NOTES

6.1 Intended use.- The meters covered by this specification are intended for use on aircraft in measuring direct currents and voltages.

6.2 Superseding data.- This specification supersedes Specification MIL-A-6752, which was the number assigned by cover sheet to Specification AN-A-19, and Specification MIL-A-5125(USAF).

6.2.1 The Office of Standardization has authorized the revision to this specification, previously fully coordinated, as an ASG limited coordination specification in view of its primarily aeronautical function.

6.3 Ordering data.- Requisitions, contracts, and orders should state the part number of the meter desired, and whether overseas packing is desired. (See Section 5.)

6.4 Provisions for Qualification tests.- In the procurement of products requiring qualification, the right is reserved to reject bids on products that have not been subjected to the required tests and found satisfactory for inclusion on the Military Qualified Products List. The attention of suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government, tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Requests for information pertaining to qualification of products covered by this specification should be addressed to the Bureau of Aeronautics, Navy Department, Washington 25, D.C., the activity responsible for qualification, with a copy to the Commander, Wright Air Development Center, Wright-Patterson Air Force Base, Ohio.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T07:24Z
Check the source to verify that this is the current version before use.

PATENT NOTICE: When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

Custodians:
Navy - Bureau of Aeronautics
Air Force

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T07:24Z
Check the source to verify that this is the current version before use.

SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)* | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT
$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

DD FORM 1 APR 64 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE PLATE NO. 15419

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T07:24Z
Check the source to verify that this is the current version before use.